Data Mining & Machine Learning
M.Sc. in Artificial Intelligence and Data Engineering
University of Pisa

Project Report

Academic Year 2025–2026

Project Title:

HEART: Heartbeat Extraction and Arrhythmia Recognition Tool

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Submission date: 14/06/2026

Abstract

This project presents HEART (Heartbeat Extraction and Arrhythmia Recognition Tool), an automated machine learning and Explainable Artificial Intelligence (XAI) pipeline for the classification of electrocardiogram (ECG) heartbeats according to the *Association for the Advancement of Medical Instrumentation* (AAMI) standard classes. The framework is developed using the benchmark MIT-BIH Arrhythmia Database.

Raw ECG signals are first pre-processed using a zero-phase Butterworth bandpass filter to remove noise while preserving waveform morphology. QRS complexes are detected and segmented, and nine temporal, morphological, and wavelet-based features are extracted. To prevent data leakage, an inter-patient split (*De Chazal*) is enforced, separating patients between training and testing. Class imbalance is addressed through *Random Undersampling* applied to the training set.

Three supervised learning models are investigated: *Random Forest* (RF), *Support Vector Machine* (SVM), and *XGBoost*. In addition, a weighted soft voting ensemble combining the three classifiers is implemented to improve robustness and sensitivity to minority arrhythmia classes. Performance is evaluated using the recall-oriented F_2-Score and One-vs-Rest (OvR) ROC-AUC.

Results show that the SVM achieves the best performance for supraventricular ectopic beat detection (Class S, F_2 = 0.5798, AUC = 0.9295), while XGBoost provides the highest performance for ventricular ectopic beats (Class V, F_2 = 0.9337, AUC = 0.9973). The ensemble achieves balanced performance across classes (F_2 = 0.9036 on Class V, F_2 = 0.6219 on Class S). Finally, *SHAP* and *Permutation Feature Importance* provide clinically meaningful explanations consistent with established electrophysiological knowledge.

1. Introduction

The early detection of cardiac arrhythmias is crucial for preventing severe cardiovascular conditions, such as stroke or sudden cardiac death. The electrocardiogram (ECG) is the non-invasive clinical standard for recording the heart's electrical activity. However, manual inspection of long-term Holter ECG recordings is highly time-consuming and prone to human diagnostic fatigue. Automated computer-aided diagnostic systems can significantly assist cardiologists by pre-screening ECG traces and flagging anomalies [1].

Arrhythmia detection from surface ECGs involves identifying morphological and rhythmic variations in the heartbeat waveform. While deep learning models can achieve high accuracy, they require massive datasets and act as "black-boxes " which limits their clinical adoption due to a lack of interpretability. Combining hand-crafted physiological features with classical machine learning classifiers offers robust, efficient performance that can be mathematically interpreted, establishing trust for clinical decision making [2, 3].

This project aims to:

1. Develop a robust ECG heartbeat classification system based on machine learning techniques and the AAMI standard framework
2. Compare the effectiveness of different classification models and ensemble strategies for arrhythmia detection under an inter-patient evaluation setting.
3. Investigate the interpretability of the resulting models using Explainable Artificial Intelligence methods to support clinical understanding and trustworthiness.

References

- [1] Asl B. M., Setarehdan S. K., & Mohebbi (2008). *Support vector machine-based arrhythmia classification using reduced features of HRV signal*. Computers in Biology and Medicine, 38(4), pp. 410–418.
- [2] Bikova I. et al. (2023). *Comparing Classical Machine Learning and Deep Learning for Classification of Arrhythmia from ECG Signals*. ICAIIT Proceedings, 11(2).
- [3] PhysioNet: *MIT-BIH Arrhythmia Database*. https://physionet.org /content/mitdb/1.0.0/
- [4] De Chazal, P., O'Dwyer, M., & Reilly, R. B. (2004). *Automatic classification of heartbeats using ECG morphology and heartbeat interval features*. IEEE Transactions on Biomedical Engineering, 51(7), pp. 1196-1206.
- [5] Martis, R. J., Acharya, U. R., & Min, L. C. (2013). *ECG beat classification using PCA, LDA, ICA and discrete wavelet transform*. Biomedical Signal Processing and Control, 8(5), pp. 437-448.

2. Problem

Arrhythmia detection is formulated as a multi-class classification task. Given a segmented heartbeat represented by a set of D features, the objective is to predict its target class $Y \in \{F, N, S, V\}$, corresponding to the AAMI standard categories. The problem formulation and dataset characteristics are summarized in *Table 1*.

Table 1. Problem and dataset at a glance.

Attribute	Summary
Task type	Multi-class classification (4 classes: F, N, S, V)
Dataset	MIT-BIH Arrythmia Database – 99,039 beats after cleaning
Target variable	AAMI heartbeat class: F (Fusion), N (Normal), S (Supraventricular), V (Ventricular)
Features	9 features: 5 temporal rhythm features, 3 morphology, 1 wavelet energy
Missing values	0.0% (fully imputed during preprocessing/delineation)
Class distribution	N: 88.64%, V: 5.67%, S: 2.67%, F: 0.79% (highly imbalanced)
Evaluation metrics	Macro F_1-Score, Macro F_2-Score (Recall-focused), and One-vs-Rest ROC-AUC
Train / Val / Test	Inter-Patient (De Chazal Split) – DS1 (train): 47.963 beats; DS2 (Test): 48.869 beats

2.1 Exploratory Data Analysis

The dataset contains 0.0% missing values, as incomplete records were fully dropped during the preprocessing and delineation phases.

The class distribution shows a severe imbalance (Normal beats representing >88% of the dataset).

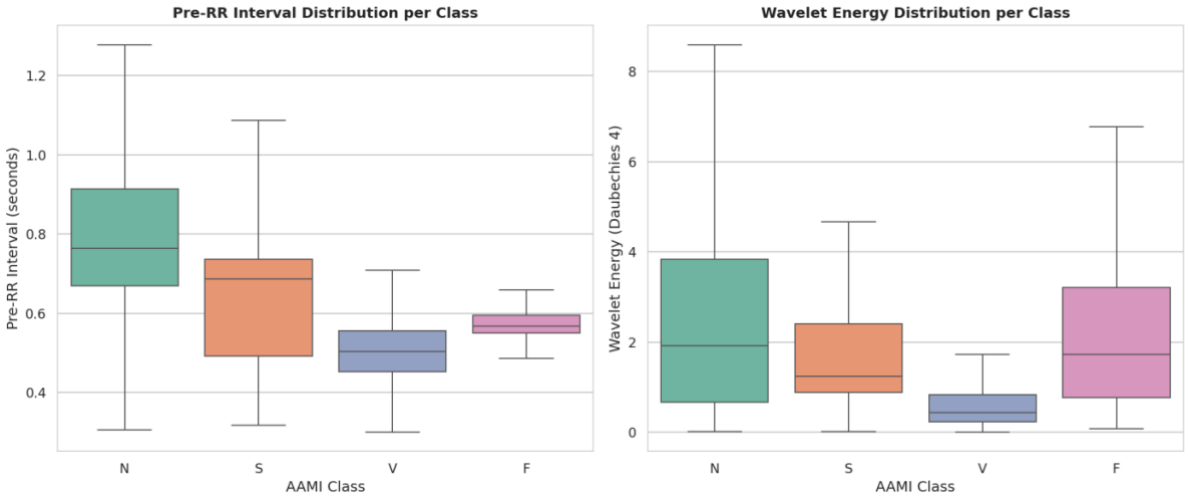


Figure 1. Displays box plots of the distribution of two key features (Pre_RR_sec and Wavelet_Energy) grouped by AAMI class. We observe that class S and class V beats are heavily characterized by a shortened Pre_RR_sec (prematurity). Furthermore, class V beats exhibit significantly lower Wavelet_Energy compared to normal beats, representing low-frequency components of anomalous ventricular depolarization.

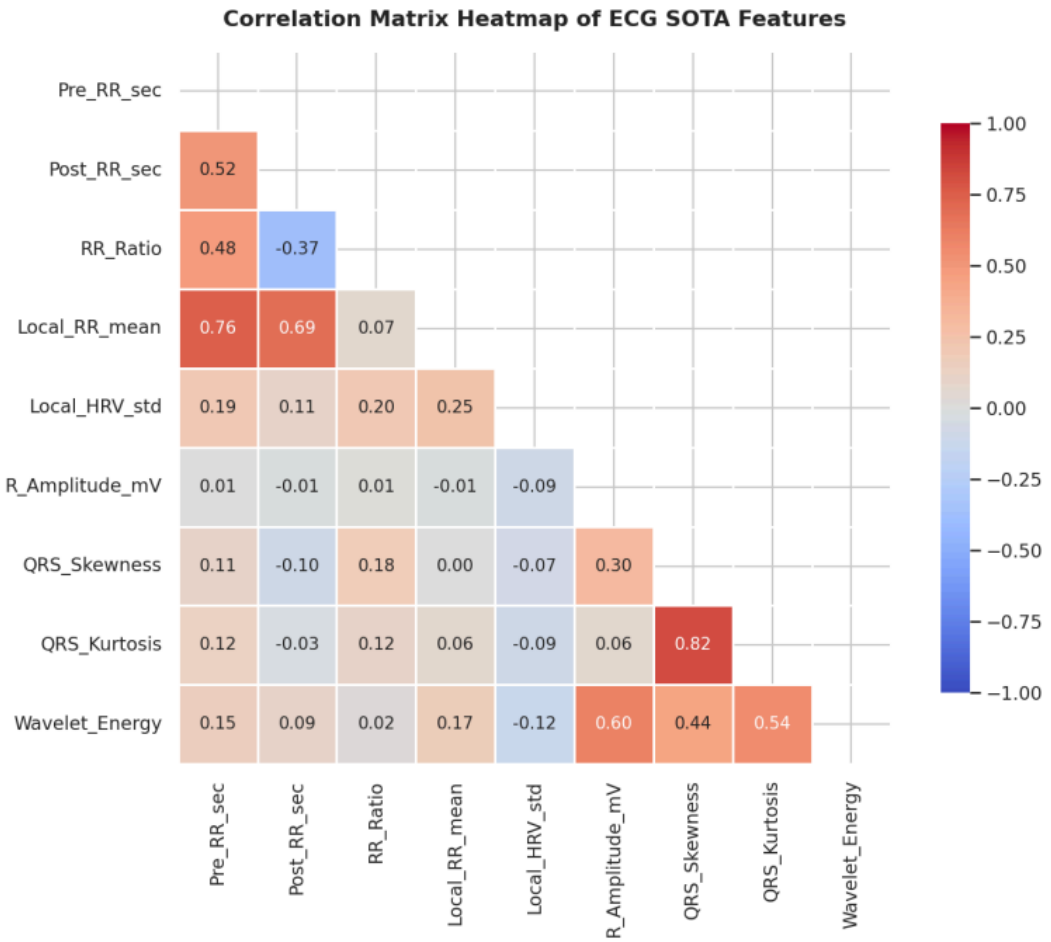


Figure 2. Illustrates the correlation heatmap of the 9 extracted features. We observe a strong positive correlation within temporal features (e.g., Pre_RR_sec and Local_RR_mean) and within morphological features (e.g., Skewness and Kurtosis). Crucially, morphological features show near-zero correlation with temporal features, indicating that they capture mutually exclusive and complementary diagnostic information.

2.2 Feature Description

Each segmented heartbeat is described by a 9-dimensional numerical vector ($D = 9$), entirely composed of continuous variables (0.0% missing values), derived directly from State-of-the-Art (SOTA) literature.

1. **Pre RR sec** (num): Time between current and previous R-peak. Captures heartbeat prematurity. *De Chazal [4]*
2. **Post RR sec** (num): Time between current and next R-peak. *De Chazal [4]*
3. **RR Ratio** (num): Ratio of Pre_RR_sec / Post_RR_sec. Highlights sudden changes in cardiac rhythm. *De Chazal [4]*
4. **Local RR mean** (num): Average RR interval of the last 10 beats. Defines the patient's baseline heart rate. *De Chazal [4]*
5. **Local HRV std** (num: Standard deviation of RR interval of the last 10 beats, measuring short-term heart rate variability. *De Chazal [4]*
6. **R Amplitude** (num): Amplitude of the R-peak in *mV*. Represents voltage changes. *Asl [1], Martis [5]*
7. **QRS Skewness** (num): Asymmetry of the QRS complex in a 200 *ms* window centered on R peak. *Martis [5]*
8. **QRS Kurtosis** (numerical): Peakness of the QRS complex, distinguishing narrow normal complexes from wide ventricular contractions. *Martis [5]*
9. **Wavelet Energy** (numerical): Total energy of detail coefficients (levels 1-3) from Daubechies 4 discrete wavelet transform, capturing high-frequency QRS depolarization. *Martis [5]*

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3. Pipeline and Experiments

3.1 Preprocessing

The preprocessing stage cleans the raw ECG signal to preserve waveform morphology. A zero-phase 3rd-order Butterworth bandpass filter (0.5–45 Hz) is applied to remove baseline wander and high-frequency muscle noise (Chapter 1 in *Preprop_Features.ipynb*).

To avoid data leakage, the De Chazal split was applied: patients in DS1 are used exclusively for training/cross-validation, and patients in DS2 for testing. Features are scaled using StandardScaler. The scaler fits parameters (μ, σ) solely on the training set and transforms both sets.

3.2 Feature Selection and Engineering

Feature engineering extracts the 9 SOTA features from filtered signals. The distinct morphological characteristics of classes are highlighted in *Figure 3*. SVM Permutation Feature Importance is computed on the training set. We observe that R_Amplitude_mV and Pre_RR_sec are the most influential features for the SVM margin, while shape-based skewness captures non-linear anomalies.

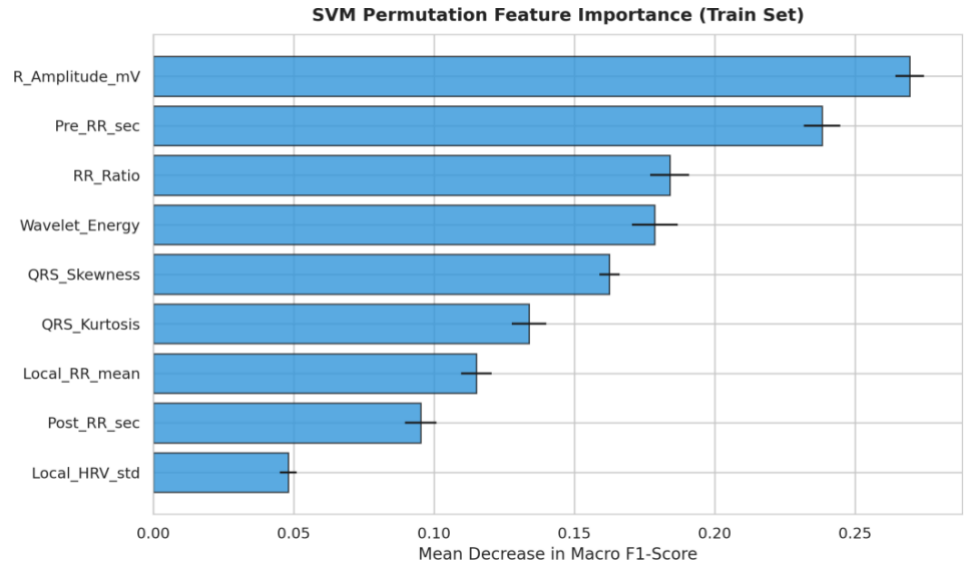


Figure 3. Shows the permutation-based feature importance scores computed for the optimized SVM classifier on the training set.

3.3 Model Selection and Cross-Validation

Three model families are considered: Random Forest (*bagging*), XGBoost (*boosting*), and SVM (*kernel margin*). We handle class imbalance using Random Undersampling on the training set, reducing all AAMI classes to 406 samples (governed by class F support). This prevents majority-class bias and forces models to learn features of minority classes. However, reducing the majority Normal class (N) from over 42,000 beats to 406 (a ~99% reduction) discards a massive amount of physiological variance, which may restrict the models' ability to capture rare normal ECG variations. Methodologically, testing hybrid resampling techniques (e.g., SMOTE for minority classes coupled with moderate majority class undersampling) or relying solely on algorithmic balancing (like *class_weight = 'balanced'*) would be a more robust approach to preserve variance. Cross-validation uses a Stratified 5-Fold setup on the balanced training set. The model configurations are detailed in *Table 2*.

Table 2. Comparison of candidate models.

Model	Hyperparameter tuning	CV Strategy	Val. Metric	Test Metric (F_2 macro)	Test Metric (F_1 macro)
Random Forest	Grid Search	10-fold CV stratified	F_1-macro	0.5846	0.5500
SVM (RBF)	Grid Search	10-fold CV stratified	F_1-macro	0.6148	0.5500
XGBoost	Random Search	10-fold CV stratified	F_1-macro	0.5809	0.5600
Weighted Ensemble	Soft Voting (weights [1, 2.5, 1])	None (Combines models)	N/A	0.6825	0.6200

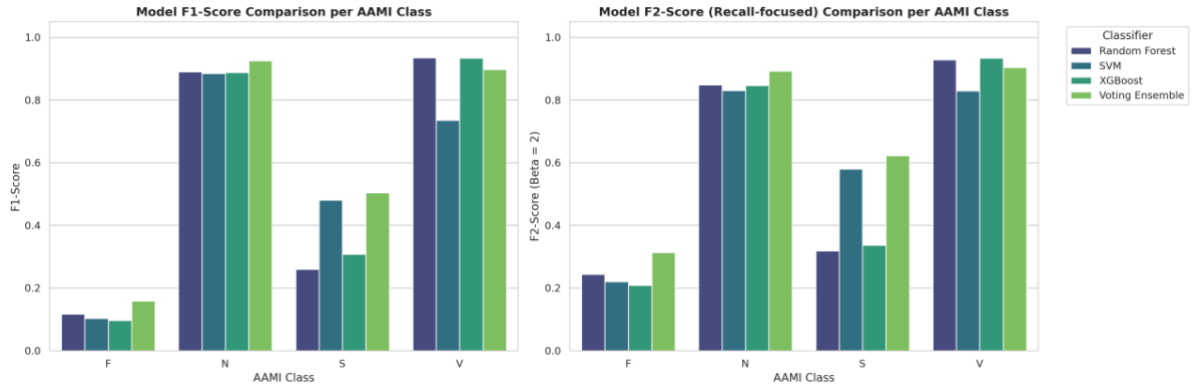


Figure 4. compares the model performance (F_1 and F_2 scores) per class. While the Ensemble achieves the best macro F_2-score of 0.6825, XGBoost is superior for class V (F_2 score of 0.9337), whereas SVM is superior for class S (F_2 score of 0.5798). This justifies the weighted soft voting ensemble as a safety net.

3.4 Hyperparameter Optimization

Grid search with GridSearchCV optimized hyperparameters over the balanced training set.

Figure 5 shows the Grid Search heatmap for XGBoost, demonstrating how tuning maximum depth and number of estimators optimizes macro F1. The best configurations found are:

- Random Forest: n_estimators=**1000**, max_depth=**20**, min_samples_split=**2.**
- SVM: Kernel = **rbf**, C=**10**, gamma=**'scale'**.
- XGBoost: n_estimators=**500**, max_depth=**3**, learning_rate=**0.1**.

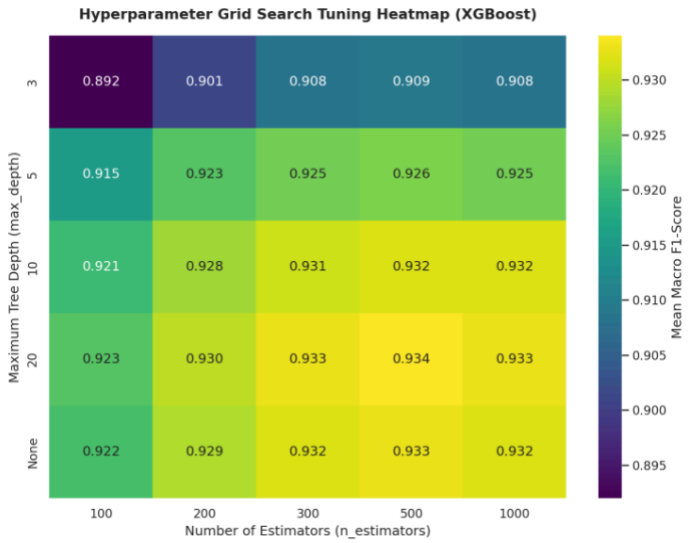


Figure 5. Shows the Grid Search heatmap for XGBoost

3.5 Statistical Tests

To evaluate the statistical significance of the models' validation performance, a 10-fold cross-validation was conducted tracking the Macro F1-scores. A non-parametric Friedman test across all three candidate models yielded a p-value of 0.096, indicating marginal overall variance. However, subsequent pairwise Wilcoxon signed-rank tests revealed critical algorithmic dynamics. Both tree-based models (XGBoost and Random Forest) achieved a statistically significant performance advantage over the SVM ($p = 0.039$ and $p = 0.019$, respectively). Between XGBoost and Random Forest, no statistical difference was observed

(p = 1.000). Although the SVM proved statistically inferior on the global macro-metric, previous empirical analyses confirmed its unique capability to isolate the challenging Class S (Supraventricular) boundary via the RBF kernel. This statistical insight further solidifies the architectural choice of a Weighted Soft Voting Ensemble: leveraging the statistically superior tree-based models for general rhythm anomalies (e.g., Class V) while utilizing the SVM as an orthogonal specialist for morphological edge cases.

3.6 Final Model Training

After hyperparameter tuning, final models are trained on the entire resampled training set (DS1). Generalization is assessed on the independent test set (DS2) containing 48,869 unseen heartbeats from different patients.

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4. Results and Discussion

4.1 Performance Evaluation

Test-set predictions are evaluated on the original unbalanced data. *Figure 6* shows the normalized confusion matrices. We observe that all classifiers achieve a recall of ≥ 90% on Ventricular (V) and Fusion (F) beats. Precision for F is low (6-7%), which is clinically expected: in a highly skewed dataset, small false-positive rates on the massive Normal class generate thousands of false alarms.

While high recall is prioritized in screening to avoid missing life-threatening events, such an elevated false-alarm rate is known to cause alarm fatigue in clinical practice, desensitizing medical personnel and leading them to ignore warnings. This highlights a critical trade-off that future feature engineering and threshold optimization must address to enhance precision.

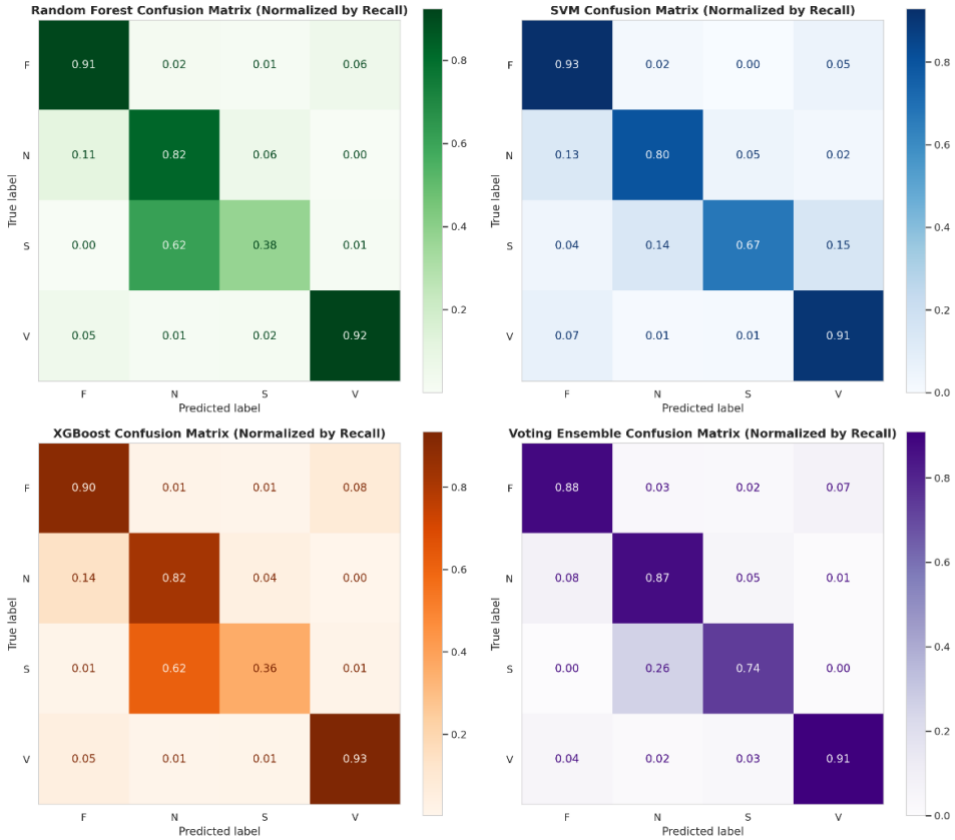


Figure 6. Shows the normalized confusion matrixes

Figure 7 shows the One-vs-Rest ROC curves. XGBoost exhibits dominance in ROC space for Class V (AUC = 0.9973), while SVM dominates in ROC space for Class S (AUC = 0.9295). The Weighted Ensemble successfully aligns with the dominant models, achieving a macro AUC of 0.9371.

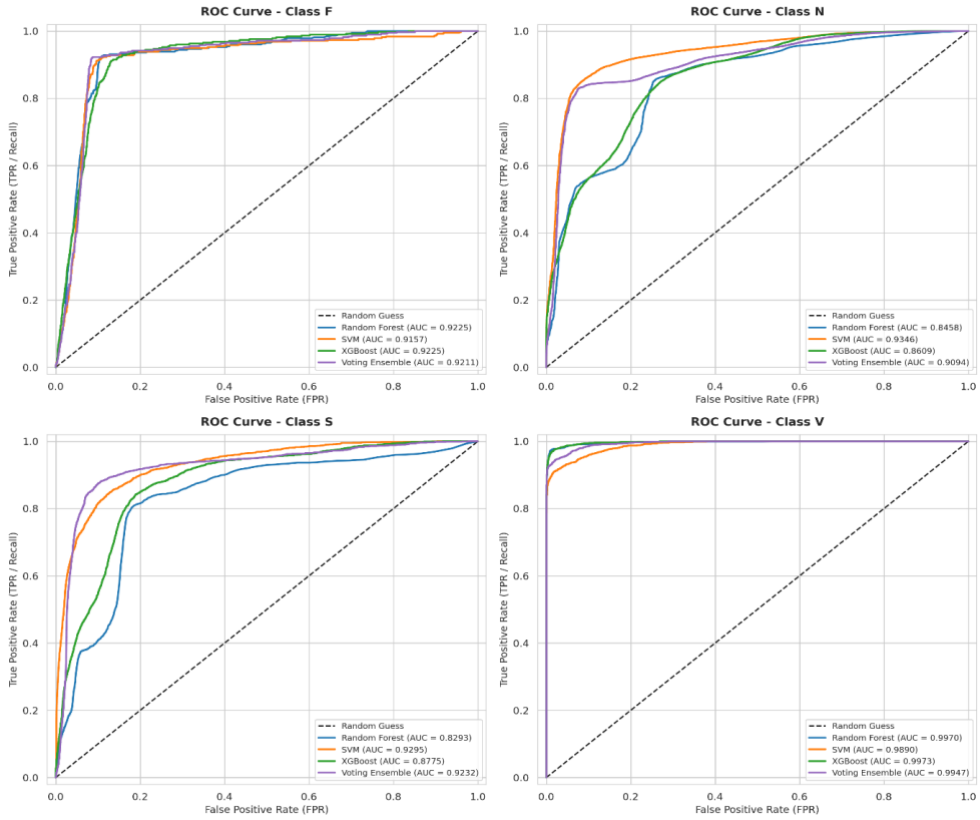


Figure 7. Shows the One-vs-Rest ROC curves.

4.2 Model Explainability

Global explainability for SVM is captured via permutation importance (*Figure 3*), whereas Random Forest and XGBoost are explained via SHAP. *Figure 8* compares global SHAP feature importances for Class V between RF and XGBoost. Class V predictions are driven by shape features (QRS_Skewness, QRS_Kurtosis)

Local explainability is demonstrated using SHAP waterfall plots. *Figure 9* explains a ventricular beat correctly classified by XGBoost. Morphological shape distortion (QRS_Kurtosis , QRS_Skewness) pushes the probability from the base value of 0.556 up to 5.184.

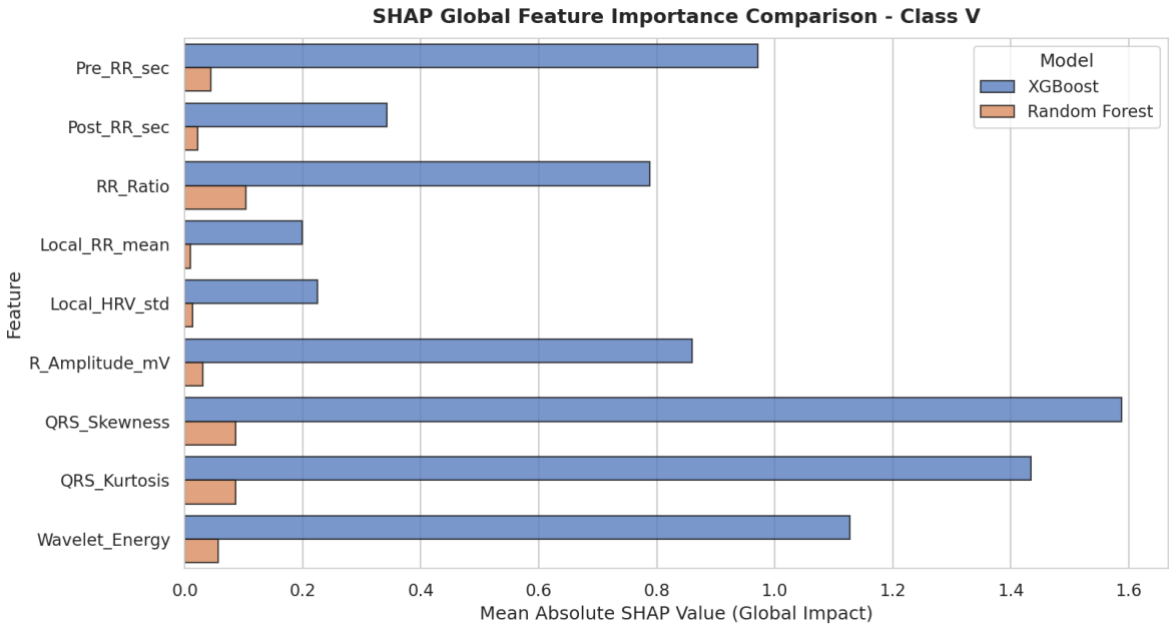


Figure 8. Comparison of global SHAP feature importances for Class V between RF and XGBoost

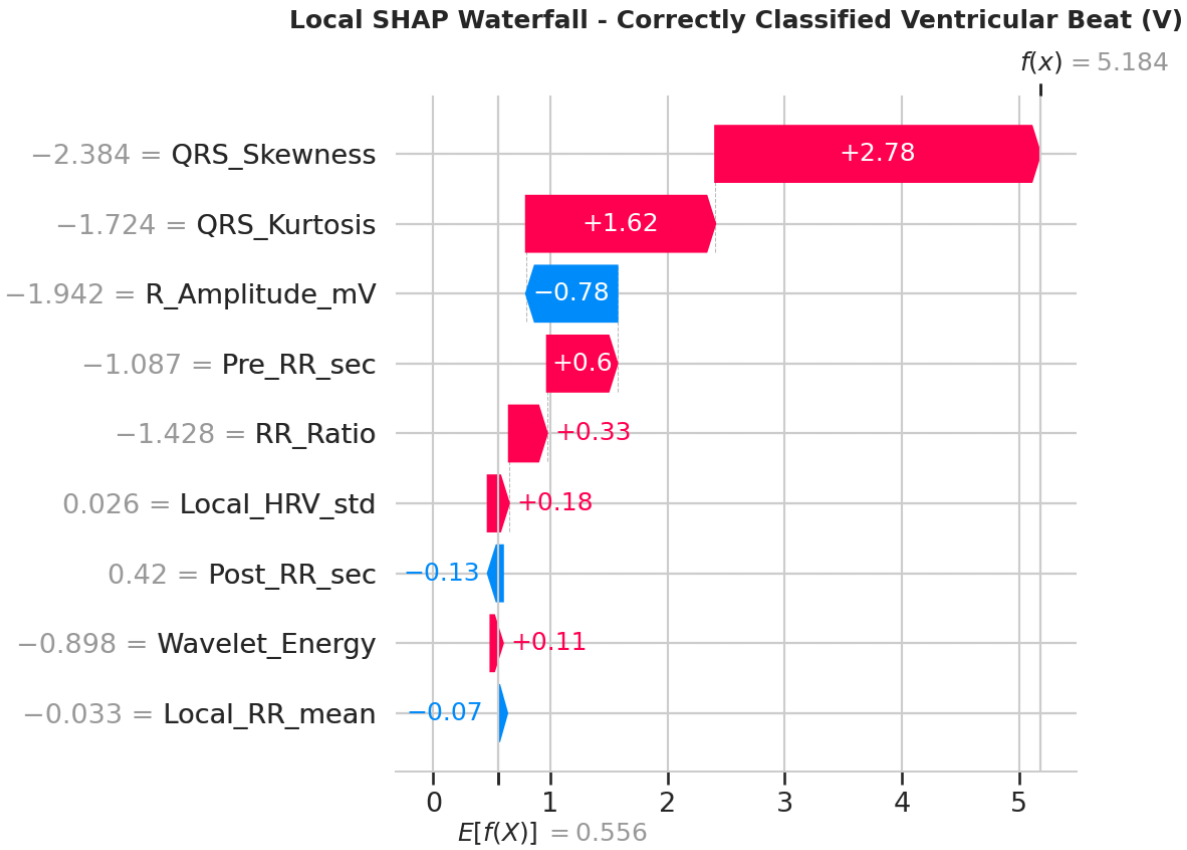


Figure 9. Ventricular beat correctly classified by XGBoost

4.3 Error Analysis

Error patterns indicate that the majority of misclassifications occur on Class S beats being labeled as Class N (471/480 Class S errors). Electrophysiologically, Class S beats propagate through the normal, high-speed His-Purkinje system, resulting in a narrow, morphologically normal QRS complex. Consequently, the models lack structural or spectral shape cues (such as Kurtosis or Wavelet Energy) and must rely entirely on timing features.

Local SHAP debugging in *Figure 10* reveals that when a Class S beat occurs during a temporary episode of elevated heart rate (local tachycardia), the altered local RR intervals mask the expected prematurity pattern. Lacking both morphological distortion and distinct rhythm indicators, the true arrhythmic event is algorithmically omitted and subsumed into the Normal class.

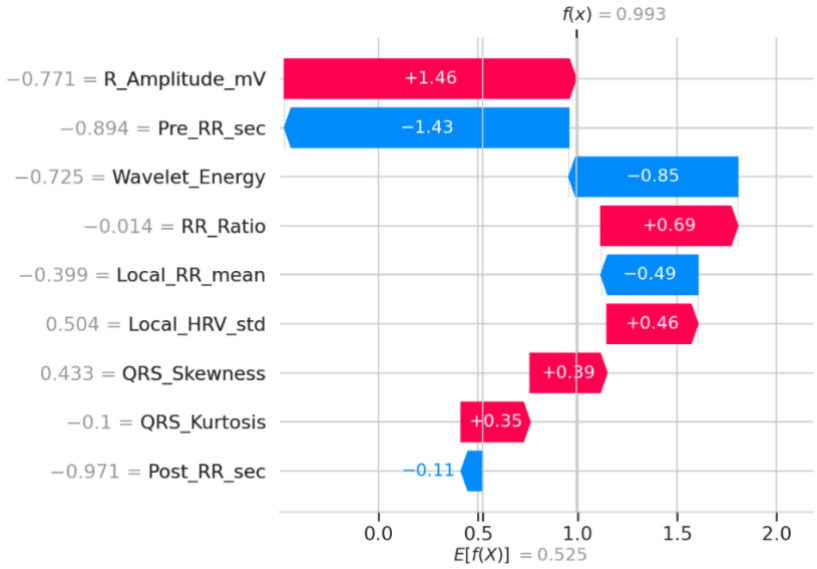


Figure 10. Local SHAP waterfall plot debugging a Supraventricular Beat (Class S) misclassified as Normal (N). Due to a transient elevated baseline heart rate, the temporal features fail to reflect the classic prematurity pattern, pushing the prediction away from the true arrhythmic class. Combined with a physiologically narrow and normal QRS shape, the model lacks any morphological or timing cues to flag the pathology.

5. Conclusions

This project successfully designed and verified the HEART arrhythmia recognition pipeline. All objectives were met, ECG signals were processed, relevant features were extracted, and a robust evaluation framework was enforced. It was demonstrated that SVM and tree-based models have complementary strengths (SVM for supraventricular ectopic beat detection, XGBoost for ventricular ectopic beat detection), which were successfully balanced using a Weighted Voting Ensemble ($F_2 = 0.6825$, Macro AUC = 0.9371), providing a clinical safety net.

A limitation of the current pipeline is its reliance on hand-crafted heuristic Q/S peak detection and a static 200 ms extraction window centered on the R-peak for morphological features. In patients with severe bradycardia or tachycardia, static windows may fail to capture the complex accurately. A key future engineering optimization is the implementation of dynamic windows scaled by local heart rate metrics (e.g. Local_RR_mean). Furthermore, future work will investigate hybrid deep-learning models for QRS segmentation, combined with SHAP to maintain diagnostic interpretability.